Exercise 2: Input, passing data to functions, and if/else

1. The opposite of a number is the number itself with its sign reversed. For example, the opposite of 1 is -1. The opposite of -6 is 6. Write some code that assigns a numerical value to a variable, and prints out the opposite of that number.

2. Write code which can convert any number of days, hours, minutes and seconds into seconds.

Can you do this as a function? Optionally, can you also write code to convert any number of seconds into days, hours, minutes and seconds?

3. Write code which asks the user for six items: two integer numbers, two decimal numbers, and two words. For each kind of input, do the following:

(a) Convert the user's integer numbers into two integers, and divide one by the other. Print the result for the user, and also print out the _type_ of the result.

(b) Convert the user's decimal numbers into two floating-point values, and multiply one by the other. Again, print the result for the user, and the type.

(c) Compare the lengths of the two words; store whether they are identical in a new variable, and use it to:

(1) Print out to the user whether the lengths are the same, and

(2) Print out the _type_ of that variable.

Try entering two integers which divide without remainder, then two which do not. What do you notice about the type of the result?

Try doing the same with the floating-point numbers. What do you notice about _their_ type?

Finally, what is the type of the variable you used to compare the two words' lengths? Why is it of that type?

4. Looking at your program from (3) above, attempt the following:

(a) Convert each word the user enters into lowercase, or uppercase. Is there any difference in the result?

(b) Try entering incorrect types of input (such as a word for an integer, and a floating-point value for a word). What happens? Can you add code which will help prevent these issues? Is it possible to cater for _every_ situation which might occur, or for all of those which might reasonably occur?